General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

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PHYSICS

1. Work done by gravity on a particle, if it moves horizontally, near the surface of earth, is
(1) Positive (2) Negative
(3) Zero (4) Both (1) and (3)

2. A force $\vec{F} = (\hat{i} - \hat{j})$N acts on a particle and displaces it from initial position $\vec{r}_1 = (3\hat{i} - 4\hat{j})$ m to final position $\vec{r}_2 = (2\hat{i} + 3\hat{j})$ m. then work done by the force is
(1) 12 J (2) –7J
(3) 6 J (4) –6 J

3. The displacement x of a body of mass 1 kg on horizontal smooth surface as a function of time t is given by $x = \frac{t}{3}$ where x is in metre and t is in seconds. The net work done in the first second is
(1) 4 J (2) 2 J
(3) 1 J (4) Zero

4. **Assertion (A):** Work done by frictional force may be negative.
Reason (R): If friction is doing work on a system it will always increase its kinetic energy.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) (A) is false but (R) is true

5. A particle moves along x axis from $x = 1$ m to $x = 3$ m under the effect of the force $F = 3x^2 - 2x + 5$ N work done in process is
(1) 24 J (2) 38 J
(3) 18 J (4) 28 J

6. Consider a system of two identical particles. One of the particles is at rest and the other has an acceleration a. Their centre of mass has an acceleration
(1) Zero (2) $\frac{a}{2}$
(3) a (4) $2a$

7. Given below are two statements:
Statement A: In a couple, total external force is zero but total torque is non-zero.
Statement B: The total torque on a system depends on the position of the origin, if the net force on system is zero.
In the light of the above statements, choose the most appropriate answer from the options given below.
(1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Statement A is correct but statement B is incorrect
(4) Statement A is incorrect but statement B is correct

8. A wheel is rotating with a speed of 420 rev/minute. After being switched off, it comes to rest in 10 s. Assuming constant angular deceleration, the number of revolutions made by it before coming to rest, will be
(1) 37 (2) 35
(3) 45 (4) 47

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9. If the position vector $(\vec{r})$ of a particle is $(\hat{i}+2\hat{j}+3\hat{k})$ m and its angular velocity $(\vec{\omega})$ is $(\hat{i}-\hat{j}+\hat{k})$ rad/s, then the linear velocity of the particle is
(1) $(2\hat{i}-2\hat{j}-3\hat{k})$ m/s (2) $(-2\hat{i}-4\hat{j}-3\hat{k})$ m/s
(3) $(-5\hat{i}-2\hat{j}-3\hat{k})$ m/s (4) $(-2\hat{i}-3\hat{j}+4\hat{k})$ m/s

10. A particle is moving parallel to x-axis with constant velocity, then about origin
(1) Torque acting on particle is constant and non-zero
(2) Angular momentum of particle is zero
(3) Angular momentum of particle is constant
(4) Angular acceleration of particle is variable

11. The potential energy of a system increases if
(1) Non-conservative forces do positive work on the system
(2) Conservative forces do positive work on the system
(3) Non-conservative forces do negative work on the system
(4) Conservative forces do negative work on the system

12. A mass m is attached to a thin wire and whirled in a vertical circle. The wire is least likely to break when
(1) The mass is at the highest point
(2) The mass is at the lowest point
(3) The wire is horizontal
(4) Inclined at an angle of 45° with the vertical

13. The potential energy of a particle at position x is given by $U = (x^3 - 4x - 2)$ J, where x is in meter. The equilibrium position of the particle will be
(1) $x = 2$ m (2) $x = 0$
(3) $x = 1$ m (4) $x = 3$ m

14. The radius of gyration a uniform rod of mass M and length l about an axis passing through mid-point of the rod and perpendicular to the length will be
(1) $\frac{l}{\sqrt{3}}$ (2) $\frac{l}{2}$
(3) $\frac{l}{2\sqrt{3}}$ (4) $\frac{l}{\sqrt{2}}$

15. Given below are two statements:
Statement I: Centre of mass of any object always coincides with centre of gravity.
Statement II: Centre of gravity is the point where total gravitational torque on the body is zero.
In the light of the above statements, choose the most appropriate answer from the options given below
(1) Both statements I and II are correct
(2) Both statements I and II are incorrect
(3) Statement I is correct but II is incorrect
(4) Statement I is incorrect but II is correct

16. A dancer on ice spin faster when she folds her arms. This is due to
(1) Increase in energy and increase in angular momentum
(2) Decrease in friction at the skates
(3) Constant angular momentum and increase in kinetic energy
(4) Increase in kinetic energy and decrease in angular momentum

17. A ball is dropped from height 10 m on the ground. If coefficient of restitution is 0.5 m, then the height of the ball after 2nd bounce is
(1) 0.625 m (2) 1.25 m
(3) 2.5 m (4) 5 m

18. An object having mass 40 kg and having velocity of 4 m/s collides perfectly inelastically with another object of mass 60 kg having velocity of 2 m/s in same direction. What is the kinetic energy lost during the process?
(1) 24 J (2) 36 J
(3) 42 J (4) 48 J

19. A motor is to be used to lift water from a depth of 10 m and discharging it on ground level at a speed 10 m s^{-1}. If 3 kg of water is lifted per second and the efficiency of motor is 60%, then minimum power of the motor should be
(1) 450 W (2) 750 W
(3) 650 W (4) 500 W

20. Power applied to a particle varies with time as $P = 3t^2 - 2t + 1$ W where t is in second. Find the change in KE between $t = 2$ s and $t = 4$ s.
(1) 22 J (2) 46 J
(3) 61 J (4) 52 J

21. A couple produces
(1) Translational motion only
(2) Rotational motion only
(3) Both translational and rotational motion
(4) Neither translational nor rotational motion

22. Moment of inertia of ring about a tangential axis perpendicular to the plane of ring will be (Assume the mass of ring to be M and radius to be R)
(1) MR^2 (2) $\frac{MR^2}{2}$
(3) $\frac{3MR^2}{2}$ (4) $2MR^2$

23. Keeping the mass of earth to be constant if its radius changes from R to $\frac{R}{\sqrt{3}}$, then new duration of the day will be
(1) 6 hours (2) 8 hours
(3) $6\sqrt{3}$ hours (4) $24\sqrt{3}$ hours

24. Two bodies of masses 4 kg and 8 kg are moving with velocities $(\hat{i}+\hat{j}-\hat{k})$ ms^{-1} and $(2\hat{i}-2\hat{j}-4\hat{k})$ ms^{-1} respectively. The velocity (in ms^{-1}) of their centre of mass is
(1) $\hat{i}-\hat{j}+7\hat{k}$ (2) $5\hat{i}-3\hat{j}-7\hat{k}$
(3) $\frac{5}{3}\hat{i}-\hat{j}-\frac{7}{3}\hat{k}$ (4) $3\hat{i}-\hat{j}-7\hat{k}$

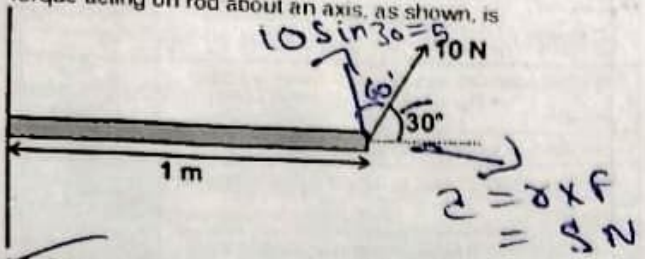
25. A : The dot product of torque and force is always zero.
R : $\vec{\tau} = \vec{r} \times \vec{F}$, so $\vec{\tau}$ and $\vec{F}$ are perpendicular to each other (symbols having usual meanings).
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion.
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion.
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements.

26. A 500 kg boat is 9 m long and is floating without motion on still water. A man of mass 100 kg is at one end and move to other end of boat and stops. The displacement of boat is
(1) 1.5 m in the direction of displacement of man
(2) 0.75 m in the direction of displacement of man
(3) 1.5 m in the direction opposite to displacement of man
(4) 0.75 m in the direction opposite to displacement of man

27. A particle is moving with constant speed of 2 m/s along the line $y = 2$m in positive x-direction. The magnitude of its angular velocity about origin when its position makes 30° with x-axis is
(1) 0.25 rad/s (2) 0.35 rad/s
(3) 0.5 rad/s (4) 0.75 rad/s

28. Torque acting on a body may be equal to (where symbols have their usual meanings)
(1) $I\alpha$
(2) $\vec{\omega}\frac{dI}{dt}$
(3) $\frac{d\vec{L}}{dt}$
(4) All of these

29. Torque acting on rod about an axis, as shown, is



(1) 5 Nm
(2) 10 Nm
(3) $5\sqrt{3}$ Nm
(4) $10\sqrt{3}$ Nm

30. **Statement A:** The virtue by which a body revolving about an axis opposes the change in rotational motion is known as moment of inertia.
Statement B: Moment of inertia depends on angular velocity, angular acceleration, torque and angular momentum.
Statement C: Moment of inertia does not depend on mass of body, mass distribution of body, position of axis of rotation.
Consider the given statements and choose the correct option.
(1) Only statement A is correct
(2) Only statements B and C are correct
(3) Only statement B is correct
(4) Only statements A and C are correct

31. Some type of collision is given in column-I and their respective co-efficient of restitution and kinetic energy in column-I. Match the collision with their respective co-efficient of restitution and energy and choose the correct option.

	Column I		Column II
(A)	Perfectly Elastic collision	(P)	$e = 0$
(B)	Inelastic collision	(Q)	$0 < e < 1$
(C)	Explosion	(R)	$e = 1$
(D)	Perfectly inelastic collision	(S)	Final K.E > Initial K.E

(1) (A) → P, (B) → Q, (C) → R, (D) → S
(2) (A) → R, (B) → P, (C) → Q, (D) → S
(3) (A) → S, (B) → R, (C) → Q, (D) → P
(4) (A) → R, (B) → Q, (C) → S, (D) → P

32. The potential energy of a conservative system is given by relation $U = ax^2 - bx$, where x represents position of the particle and a and b are constant. What is the force acting on the system?
(1) $2ax - b$
(2) $ax + b$
(3) $-2ax + b$
(4) $ax^3 + bx^2$

33. A simple pendulum of length 'l' has a bob of mass 'm'. If it is released from horizontal position, then velocity of the bob at lowest position will be (Symbols have their usual meaning)
(1) $\sqrt{3gl}$
(2) $\sqrt{2gl}$
(3) $\sqrt{5gl}$
(4) $\sqrt{gl}$

34. A conservative force of $(-3\hat{i})$ N acts on a body of mass 3 kg. The change in potential energy of the body due to the force, when it undergoes a displacement $6\hat{i}$ m is
(1) 21 J
(2) 18 J
(3) 19 J
(4) 15 J

35. An angular impulse J is applied to a hollow cylinder of mass 3 kg and radius 25 cm rotating about its axis. If the change in its angular speed is 16 rad/s, then J is (in N m s)
(1) 27
(2) 9
(3) 3
(4) 1

36. The moment of inertia of a rotating body increases from I to $\frac{4}{3}I$ without any external torque. The percentage change in its angular speed is
(1) 20%, increases
(2) 20%, decreases
(3) 25%, increases
(4) 25%, decreases

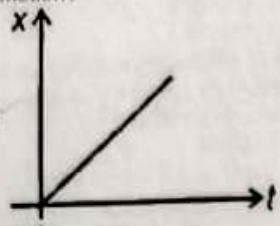
37. Work done by frictional force
(1) Always negative
(2) Always positive
(3) May be positive, negative or zero
(4) Is zero

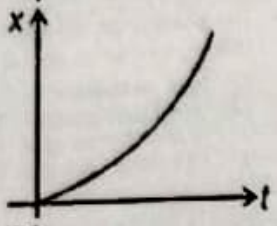
38. A body of mass 8 kg at rest explodes into three pieces, two of which of mass 2 kg each are thrown off in perpendicular directions with velocities of 1 ms^{-1} and $\sqrt{3}$ ms^{-1} respectively. The kinetic energy of third mass is
(1) 4 J
(2) 8 J
(3) 2 J
(4) 6 J

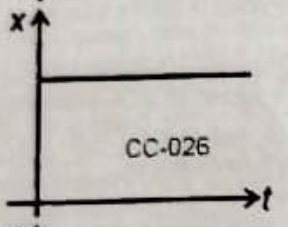
39. A person lifts a ball of mass 1 kg through the height of 1 m in 10 s. The average power supplied by him is [Take g = 10 ms^{-2}]
(1) 10 W
(2) 1 W
(3) 0.1 W
(4) 2 W

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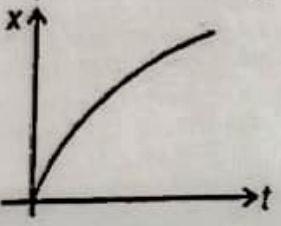
40. A body is moving unidirectionally under the influence of a source of constant power. Which of the diagrams shown below correctly shows the displacement-time (x-t) curve for its motion?

(1) 

(2) 

(3) 

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(4) 

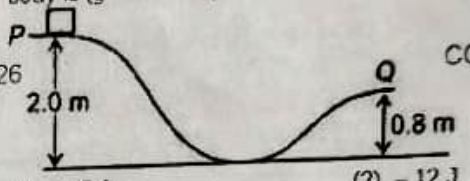
41. A particle of mass m is attached with a vertical string of length l and is given a velocity at its lower most position such that it just complete the circle. On basis of given information match the column I with column II and choose the correct option

	Column I		Column II
a	Radial acceleration at lower most position (in m/s^2)	(i)	g
b	Tension in string at horizontal position	(ii)	Zero
c	Radial acceleration at upper most position (in m/s^2)	(iii)	$3\ mg$
d	Tension in string at uppermost position (in N)	(iv)	$5\ g$

(1) a(ii), b(i), c(iv), d(iii)
(2) a(iv), b(iii), c(i), d(ii)
(3) a(i), b(iv), c(iii), d(ii)
(4) a(i), b(iii), c(ii), d(iv)

42. A body of mass 1 kg is placed at the point P on a rough track shown in figure. If slightly pushed towards right it stops at point Q of the track. The work done by frictional force on the body is (g = 10 m/s^2)

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(1) – 15 J
(2) – 12 J
(3) – 8 J
(4) – 20 J

43. The direction of angular velocity vector is
(1) Radially outward
(2) Radially inward
(3) Along the tangent to the circular path
(4) Along the axis of rotation

44. CGS unit of work is
(1) Slug
(2) Dyne
(3) Erg
(4) Joule

45. A ball moving on a smooth horizontal surface with velocity $\left(2\hat{i}+2\hat{j}\right)$ strikes a smooth wall in (y - z) plane. The coefficient of restitution between ball and wall is $\left(\frac{1}{2}\right)$. The velocity of ball, just after it hits the wall is
(1) $\hat{i}-\hat{j}$
(2) $-\hat{i}+2\hat{j}$
(3) $-\hat{i}-2\hat{j}$
(4) $2\hat{i}-\hat{j}$

CHEMISTRY

46. Match List-I with List-II

	List-I (Species)		List-II (Bond angle)
(a)	BF_3	(i)	Contains only 180° and 90° bond angle
(b)	XeF_4	(ii)	Contains only 120° bond angle
(c)	PCl_5	(iii)	Contains 90°, 180° and 120° bond angles
(d)	XeF_2	(iv)	Contains only 180° bond angle

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The correct match is
(1) a(ii), b(iii), c(i), d(iv)
(2) a(i), b(iii), c(ii), d(iv)
(3) a(ii), b(i), c(iii), d(iv)
(4) a(i), b(ii), c(iv), d(iii)

47. Among CaH_2, BeH_2, BaH_2, the order of ionic character is
(1) $BeH_2 < CaH_2 < BaH_2$
(2) $CaH_2 < BeH_2 < BaH_2$
(3) $BaH_2 < BeH_2 < CaH_2$
(4) $BeH_2 < BaH_2 < CaH_2$

48. In the structure of ClF_3, the number of lone pair of electrons on central atom 'Cl' is
(1) One
(2) Two
(3) Three
(4) Four

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49. Which among the following species has $p\pi-p\pi$ overlapping?
(1) H_2
(2) CO_2
(3) BeH_2
(4) H_2O

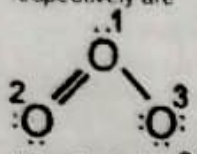
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50. The incorrect statement among the following with regards to resonance is
(1) Resonance averages the bonds characteristics as a whole
(2) The canonical forms have no real existence
(3) There is an equilibrium between canonical forms
(4) Resonance stabilises the molecules as the energy of resonance hybrid is less than energy of any canonical structures

51. Maximum dipole moment among the following is of
(1) NF_3 (2) NH_3
(3) BF_3 (4) CCl_4

52. In which of the following processes, the stability increases and the magnetic behaviour changes?
(1) $Li_2 \rightarrow Li_2^+$ (2) $O_2 \rightarrow O_2^+$
(3) $N_2^+ \rightarrow N_2$ (4) $H_2 \rightarrow H_2^+$

53. The formal charges of oxygen labelled 3,2,1 in ozone respectively are



(1) −1, +1, 0 (2) 0, 0, −1
(3) −1, 0, +1 (4) 0, +1, −1

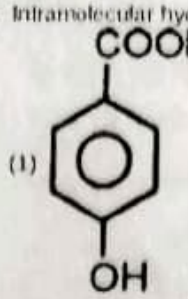
54. The correct order of bond length of C – C, C – N, C – H is
(1) C – H < C – N < C – C (2) C – C < C – N < C – H
(3) C – C < C – H < C – N (4) C – N < C – H < C – C

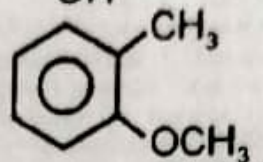
55. Consider the following species:
(a) PF_5
(b) SF_6
(c) H_2SO_4
(d) NO_2
(e) CCl_4
The expanded octet is observed in
(1) (c), (d) and (e) only (2) (a), (b) and (c) only
(3) (b), (c) and (d) only (4) (a), (b) and (e) only

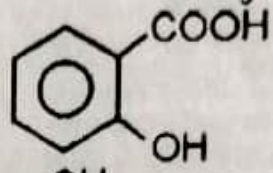
56. Cl–O bond order in ClO_4^- ion is
(1) 1.25 (2) 1.50
(3) 1.66 (4) 1.75

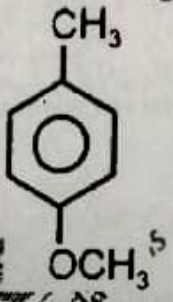
57. The number of covalent bonds present in CaC_2 is/are
(1) 4 (2) 3
(3) 2 (4) 1

58. Intramolecular hydrogen bond is present in

(1) 

(2) 

(3) 

(4) 

59. Which of the following molecules has an odd electron?
(1) H_2O (2) CH_4
(3) NH_3 (4) NO

60. Identify a molecule which does not exist.
(1) Li_2 (2) C_2
(3) O_2 (4) He_2

61. Given below are two statements one is labelled as assertion (A) other is labelled as reason (R).
Assertion (A): O_2 is a paramagnetic species.
Reason (R): O_2 has two unpaired electrons in π-antibonding molecular orbital.
In the light of the above statements, choose the correct answer from the options given below.
(1) Both A and R are true and R is the correct explanation of A
(2) Both A and R are true but R is NOT the correct explanation of A
(3) A is true but R is false
(4) A is false but R is true

62. Match List-I with List-II

	List-I (Molecule)		List-II (Shape)
a.	ClF_3	(i)	Bent
b.	SO_2	(ii)	See-saw
c.	SF_4	(iii)	Trigonal Pyramidal
d.	NF_3	(iv)	T-shape

The correct match is
(1) a(ii), b(iii), c(i), d(iv) (2) a(iv), b(i), c(iii), d(ii)
(3) a(iv), b(i), c(ii), d(iii) (4) a(i), b(iv), c(ii), d(iii)

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63. Given below are the two statements:
Statement I: Octet theory does not account for the shape of the molecules.
Statement II: Octet theory is not applicable to SCl_2 molecule.
In the light of above statements, choose the correct answer from options given below.
(1) Statement I is correct but statement II is incorrect
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and statement II are correct
(4) Both statement I and statement II are incorrect

64. If the Dipole moment of AB molecule is 2.5 D and A–B bond length is 1.5 Å then the percentage ionic character of AB molecule will be
(1) 24.2%
(2) 52.1%
(3) 18.4%
(4) 34.7%

65. Correct order of C–O bond length of CO, CO_2 and CO_3^{2-} is
(1) $CO_2 > CO_3^{2-} > CO$
(2) $CO_3^{2-} > CO > CO_2$
(3) $CO > CO_2 > CO_3^{2-}$
(4) $CO_3^{2-} > CO_2 > CO$

66. Decreasing order of O–O bond length in the given species is
(1) $O_2^- > O_2^- > O_2$
(2) $O_2 > O_2^+ > O_2^-$
(3) $O_2^+ > O_2 > O_2^-$
(4) $O_2^- > O_2 > O_2^+$

67. Among of the following ionic compounds, the one that has maximum lattice enthalpy is
(1) NaF
(2) NaCl
(3) LiCl
(4) LiF

68. The hybridization of atomic orbitals of xenon in XeO_3, XeF_4 and XeF_2 respectively are
(1) sp^2, sp^3d^2 and sp^3d
(2) sp^3, sp^3d and sp^3d^2
(3) sp^3, sp^3 and sp
(4) sp^3, sp^3d^2 and sp^3d

69. Consider the following statements:
(a) Methyl alcohol is soluble in water.
(b) Boiling point of HCl is higher than HF.
(c) Magnitude of H-bonding is maximum in solid state and minimum in the gaseous state.
(d) The hydrogen bonds have strong influence on the structure and properties of the compounds.
The correct statements are
(1) (a) and (b) only
(2) (a), (b) and (c) only
(3) (a), (c) and (d) only
(4) (a), (b), (c) and (d)

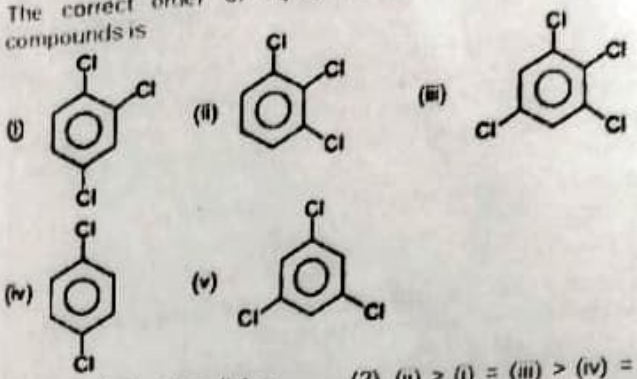
70. Which among the following is lowest unoccupied molecular orbital of N_2 ?
(1) σ^*2s
(2) π^*2p_x
(3) σ^*2p_z
(4) $\sigma 2p_z$

71. Maximum number of atoms present in a plane in SbF_5 is
(1) 5
(2) 4
(3) 3
(4) 2

72. Which of the following species is diamagnetic?
(1) O_2
(2) O_2^{2-}
(3) NO
(4) N_2^+

73. Polar molecule among the following is
(1) PF_3Cl_2
(2) PCl_3F_2
(3) XeO_4
(4) PCl_5

74. The correct order of dipole moment of the following compounds is



(1) (iii) > (ii) > (i) > (iv) > (v)
(2) (ii) > (i) = (iii) > (iv) = (v)
(3) (iii) > (i) > (ii) = (v) > (iv)
(4) (iii) > (i) > (iv) > (ii) > (v)

75. Shape of $N(SiH_3)_3$ and $N(CH_3)_3$ will be respectively
(1) Planar, pyramidal
(2) Pyramidal, planar
(3) Planar, planar
(4) Pyramidal, pyramidal

76. Consider the following statements about hybridisation
(a) The orbitals present in the valence shell of the atom are hybridised.
(b) Promotion of electron is an essential condition prior to hybridisation.
(c) The hybrid orbitals are more effective in forming stable bonds than the pure atomic orbitals.
(d) The hybridised orbitals are always equivalent in energy and shape.
Choose the correct statements
(1) (a), (b) and (c) only
(2) (b), (c) and (d) only
(3) (a), (c) and (d) only
(4) (a), (b), (c) and (d)

77. Arrange the following compounds in correct order of covalent character
(1) $AlCl_3 < MgCl_2 < NaCl$
(2) $NaCl < AlCl_3 < MgCl_2$
(3) $NaCl < MgCl_2 < AlCl_3$
(4) $MgCl_2 < AlCl_3 < NaCl$

78. According to VSEPR theory, the repulsive interaction of electron pairs decreases in the order of
(1) Lone pair – Lone pair > Bond pair – Bond pair > Bond pair – Lone pair
(2) Bond pair – Bond pair > Bond pair – Lone pair > Lone pair – Lone pair
(3) Lone pair – Lone pair > Lone pair – Bond pair > Bond pair – Bond pair
(4) Bond pair – Lone pair > Lone pair – Lone pair > Bond pair – Bond pair

79. In the change NO^+ to NO, the electron is added to
(1) σ-orbital
(2) π-orbital
(3) σ^*-orbital
(4) π^*-orbital

80. Maximum number of lone pairs of electrons are present in
(1) XeO_2F_2
(2) IF_3
(3) BF_3
(4) XeF_4

81. Consider the following statements:
Statement I: The energy of $\sigma 2p_z$ molecular orbital is higher than that of $\pi 2p_x$ molecular orbital for F_2
Statement II: The energy of $\sigma^* 2p_z$ molecular orbital is higher than that of $\pi^* 2p_x$ molecular orbital for N_2.
In the light of above statements, choose the correct option.
(1) Both statement I and statement II are correct.
(2) Both statement I and statement II are incorrect.
(3) Statement I is incorrect while statement II is correct.
(4) Statement I is correct while statement II is incorrect.

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82. $d\pi - p\pi$ bond present in
(1) CO_3^{2-} (2) PO_4^{3-}
(3) NO_3^- (4) NO_2

83. Considering x axis to be the internuclear axis, the type of overlapping of atomic orbitals that will form π bond will be
(1) $p_x + p_x$ (2) $s + p_y$
(3) $p_z + p_z$ (4) $s + p_x$

84. In which of the following molecules all covalent, ionic and co-ordinate bond(s) are present?
(1) CaC_2 (2) NH_4NO_3
(3) N_2H_4 (4) NaCl

85. Given below are two statements. One is labelled as **Assertion (A)** and other is labelled as **Reason (R)**
Assertion (A): An atomic orbital is monocentric while a molecular orbital is polycentric.
Reason (R): An electron in an atomic orbital is influenced by one nucleus, in a molecular orbital it is influenced by two or more nuclei depending upon the number of atoms in a molecule.
In the light of the above statements, choose the correct answer from the options given below
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

86. Benzoic acid exists as dimer in benzene due to
(1) Presence of carboxyl group
(2) Intermolecular H-bonding
(3) Intramolecular H-bonding
(4) Presence of H atom at α - carbon

87.  This molecular orbital is representing
(1) $\sigma 2p_z$ orbital (2) $\pi^* 2p_x$ orbital
(3) $\sigma^* 2p_z$ orbital (4) $\pi 2p_x$ orbital

88. Given below are two statements
Statement I : There are six hybrid orbitals in sp^3d^2 hybridisation.
Statement II : Number of hybrid orbitals are half of the number of atomic orbitals involved in hybridisation.
In light of above statements mark the correct statements
(1) Both the statement I and II are correct
(2) Both the statement I and II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

89. The number of 109.5° bond angles in CCl_4 is
(1) Four (2) Six
(3) Three (4) Eight

90. The species which is isostructural with PH_3 is
(1) ClF_3 (2) BCl_3
(3) SO_3 (4) XeO_3

BOTANY

91. Archaebacteria differ from eubacteria in
(1) Having cellular body organisation
(2) Having 70S ribosomes
(3) Having different cell wall structure/composition
(4) Having nucleolus

92. Cellular organisation is **not** seen in
(1) Multicellular organisms
(2) Individuals with complex body organisation
(3) Prokaryotes
(4) Viruses

93. Read the following Assertion (A) and Reason (R) statements and select the **correct** option.
Assertion (A): Deuteromycetes are commonly known as imperfect fungi
Reason (R): Sexual reproduction is absent in the members of Deuteromycetes.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) Both (A) and (R) are false
(4) (A) is true but (R) is false

94. Identify the **correct** statement w.r.t. the kingdom Monera.
(1) Its sole members are grouped under a total of three categories on the basis of their shapes
(2) All the members are characterised by the presence of a rigid cell wall
(3) Vast majority of its members depend on other organisms or dead organic matter for their nourishment.
(4) Its members play a very little role in the recycling of nutrients.

95. Most notorious protozoan causing malaria is
(1) *Trypanosoma* (2) *Paramoecium*
(3) *Amoeba* (4) *Plasmodium*

96. Arrange the following steps, in **correct** order of their occurrence during sexual reproduction in the fungal class of mushroom and puffballs.
(A). Formation of a dikaryotic hyphae that gives rise to the sexual spore bearing structure.
(B). Fusion of somatic cells of different genotypes.
(C). Fusion of nuclei within the spore bearing structure.
(D). Formation of exogenously produced meiospore.
(E). Maturation and rupturing of basidiocarp.
Choose the correct option.
(1) E → B → D → C → A (2) A → B → C → D → E
(3) B → A → C → D → E (4) D → E → A → B → C

97. According to kingdom system of classification proposed by Linnaeus position of fungi is in
(1) Monera (2) Protista
(3) Archaea (4) Plantae

98. Match the column I with column II and select the **correct** option.

Column I	Column II
(a) First discovered in tobacco plant in 1892	(i) Usually contains double stranded DNA as the genetic material
(b) An infectious agent, discovered by Diener in 1971	(ii) Cannot grow in SO_2 rich environment
(c) Comprises of a head, collar, sheath, tail fibres and remains inert outside the host	(iii) Can cause leaf curling, vein clearing and dwarfing in plants
(d) A structure comprising of two different organisms living together	(iv) Are free RNA of low molecular weight

(1) a-iv, b-i, c-iii, d-ii (2) a-iii, b-iv, c-i, d-ii
(3) a-i, b-iv, c-ii, d-iii (4) a-iv, b-iii, c-i, d-ii

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99. The following figure is of a fungus



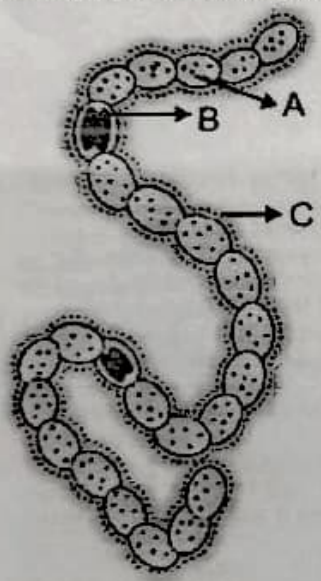
Which of the given statements is **not** true for this fungus?
(1) It is commonly known as a sac-fungus
(2) It produces conidia endogenously
(3) Its mycelium is branched and septate
(4) It forms fruiting body called ascocarp

100. Mark the **incorrect** option w.r.t. bacteria
(1) Bacterial cell structure is very complex but it has simple behaviour
(2) They show most extensive metabolic diversity
(3) They may be photosynthetic autotrophic or chemosynthetic autotrophic
(4) Ribosomes are 70S type

101. The basidiocarps are edible in
(1) Truffles (2) Morels
(3) *Agaricus* (4) Yeast

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102. Examine the figure given below and select the **correct** option giving all the three parts (A, B & C).



(1) A - Site of CO_2 fixation
B - Site of CO_2 fixation
C - Gelatinous sheath
(2) A - Site of CO_2 fixation
B - Site of N_2 fixation
C - Mucilaginous sheath
(3) A - Site of N_2 fixation
B - Site of CO_2 fixation
C - Gelatinous sheath
(4) A - Site of NO_2^- oxidation
B - Site of CO_2 fixation
C - Cell wall

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103. Fruiting bodies are **not** seen in
(1) *Agaricus* (2) Morels
(3) *Saccharomyces* (4) Truffles

104. Sleeping sickness causing protozoan is parasitic in nature. It
(1) Is ciliated
(2) Has pseudopodia
(3) Is flagellated
(4) Lacks locomotory structure

105. Identify the organism which undergoes rapid multiplication and make the sea appear red
(1) *Euglena* (2) Bladderwort
(3) *Gonyaulax* (4) *Cuscuta*

106. Bacteria do not have
(1) 70 S ribosomes
(2) Nucleoid
(3) Membrane bound cell organelles
(4) Flagella

107. Sexual reproduction is absent in
(1) Puffballs (2) *Colletotrichum*
(3) *Albugo* (4) Yeast

108. The chief producers of the oceans are
(1) Archaebacteria (2) Cyanobacteria
(3) Euglenoids (4) Diatoms

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109. One of the most notable disease caused by prions is
(1) Potato spindle tuber disease
(2) Bovine spongiform encephalopathy
(3) Mumps
(4) Influenza

110. Primitive bacteria called methanogens
(1) Are photoautotrophic bacteria
(2) Have peptidoglycan as major component in their cell wall
(3) Are abundant in marshy areas
(4) Show aerobic respiration

111. Read the following statements and select the **correct** ones.
A. *Nostoc* and *Anabaena* are photosynthetic protists.
B. The smallest living organisms lack cell wall.
C. In yeast karyogamy is delayed after plasmogamy
D. All fungi possess a purely cellulosic cell wall
(1) A and B (2) B and C
(3) C and D (4) A and D

112. Siliceous cell wall is found in
(1) Diatoms (2) Dinoflagellates
(3) Euglenoids (4) Cyanobacteria

113. Mycorrhiza is an association between
(1) Roots of higher plants and fungi
(2) Algae and fungi
(3) Roots of higher plants and algae
(4) Algae and cyanobacteria

114. The spherical shaped bacterial cell is called
(1) *Bacillus* (2) Coccus
(3) *Spirillum* (4) *Vibrio*

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115. The viruses that infect the bacteria usually have genetic material as
(1) dsDNA (2) dsRNA
(3) ssRNA (4) ssDNA

116. Mixotrophic mode of nutrition can be seen in
(1) Dinoflagellates (2) Euglenoids
(3) Protozoans (4) Chrysophytes

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117. Read the statements A and B and select the correct option.
Statement A: All the unicellular organisms are placed in kingdom Protista
Statement B: Kingdom Protista has brought together unicellular photosynthetic organisms like *Chlamydomonas*, *Chlorella*, *Spirogyra* with *Paramoecium* and *Amoeba*.
In the light of above statements, choose the **most appropriate** answer from the options given below:
(1) Statement A is correct but statement B is incorrect
(2) Statement A is incorrect but statement B is correct
(3) Both statement A and statement B are correct
(4) Both statement A and statement B are incorrect

118. The parasitic fungi on mustard is
(1) *Mucor* (2) *Rhizopus*
(3) *Albugo* (4) *Neurospora*

119. Rust fungus of wheat is
(1) *Penicillium* (2) *Mucor*
(3) *Puccinia* (4) *Rhizopus*

120. Who first proposed the biological classification of plants and animals on the basis of simple morphological characters?
(1) Aristotle (2) Linnaeus
(3) George Bentham (4) Dalton Hooker

121. Select the **incorrectly** matched pair

(1)	Viroids	–	Lack protein coat
(2)	Prions	–	Larger in size than viruses
(3)	Viruses	–	Are obligate parasites
(4)	Lichens	–	Represent symbiotic association

(1) (1) (2) (2)
(3) (3) (4) (4)

122. A zygospore is
(1) Flagellated, motile asexual spore
(2) Resting, non-motile, diploid spore
(3) Non flagellated, non-motile asexual spore
(4) Flagellated sexual spore of sac fungi

123. Mark the **correct** option for ascospores.
(a) Haploid
(b) Meiospores
(c) Exogenous
(d) Flagellated
(1) (a) & (b) (2) (b) & (c)
(3) (c) & (d) (4) (a) & (d)

124. Chlorophyllous organisms evolving oxygen in photosynthetic process are included in total how many kingdoms of Whittaker's classification system?
(1) One (2) Two
(3) Three (4) Four

125. In three domain system of classification, which kingdom is divided into two domains?
(1) Protista (2) Monera
(3) Fungi (4) Plantae

126. Read the following statements
Statement A: Chemosynthetic autotrophic bacteria oxidise various inorganic substances and use the released energy for their ATP production.
Statement B: Majority of heterotrophic bacteria are important decomposers.
In the light of above statements mark the **correct** option.
(1) Both the statements A and B are correct
(2) Both the statements A and B are incorrect
(3) Only statement B is correct
(4) Only statement A is correct

127. Fill in the blanks by selecting the **correct** option
(i) A bacteria are helpful in making curd from milk
(ii) B bacteria play a great role in recycling of nutrients like nitrogen, iron, sulphur and phosphorus
(1) A - Autotrophic, B - Chemoautotrophic
(2) A - Heterotrophic, B - Photoautotrophic
(3) A - Chemoautotrophic, B - Heterotrophic
(4) A - Heterotrophic, B - Chemoautotrophic

128. Given below features (a-c) are related to
(a) Formation of fruiting body.
(b) Cell wall around spore.
(c) Plasmodium is without cell wall.
(1) Euglenoids (2) Slime moulds
(3) Diatoms (4) Dinoflagellates

129. Ciliated protozoans differ from amoeboid and flagellated protozoans in having
(1) Locomotory structure
(2) Single celled body
(3) Heterotrophic mode of nutrition
(4) Definite regions of ingestion and egestion

130. Mark the **incorrect** statements w.r.t. euglenoids.
(1) Majority of them are found in stagnant fresh water
(2) Do not have cell wall
(3) They have one long flagellum only
(4) The pigments are identical to those present in higher plants

131. Members regarded as 'fungi imperfecti' have all of the following characteristics, **except**
(1) Decomposition of litter
(2) Formation of conidia
(3) Aseptate and branched mycelium
(4) Help in mineral recycling

132. Which of the given statements is **not** true for viruses?
(1) They are nucleoproteins where protein is infectious in nature
(2) They can be crystallised and crystals consist largely of proteins
(3) Virus means venom or poisonous fluid
(4) A virus can never have both DNA and RNA together as its genetic material

133. Read the following statements and select the **correct** option.
Statement A: In members of Phycomycetes, asexual reproduction takes place by zoospores or by aplanospores.
Statement B: Members of Ascomycetes can be saprophytic, decomposers, parasitic or coprophilous.
(1) Only statement A is correct
(2) Only statement B is correct
(3) Both statements A & B are correct
(4) Both statements A & B are incorrect

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134. In which of the following pairs both the fungi have an intervening dikaryotic stage when they reproduce sexually?
(1) *Neurospora* and *Rhizopus*
(2) *Rhizopus* and *Ustilago*
(3) *Agaricus* and *Aspergillus*
(4) *Mucor* and *Aspergillus*

135. The causal agent of 'mad cow disease'
(1) Contains DNA as genetic material
(2) Are abnormally folded proteins
(3) Are larger than viruses
(4) Do not cause any disease in humans

ZOOLOGY

136. The role of which of the following gases is quite insignificant on the respiratory rhythm?
(1) Carbon monoxide (2) Oxygen
(3) Carbon dioxide (4) Both CO and CO_2

137. Blood protein that mainly helps in maintaining osmotic balance is
(1) Fibrinogen (2) Globulin
(3) Albumin (4) Prothrombin

138. The factor in humans which differentiates serum from blood plasma is that the former lacks____. Choose the option which fills the blank **correctly**
(1) Globulins (2) Antibodies
(3) Albumins (4) Clotting factors

139. After holding the breath for a while, we feel the urge to breath due to the
(1) Increase in concentration of oxygen
(2) Increase in the pH of blood
(3) Increase in the carbon dioxide concentration
(4) Decrease in H^+ concentration

140. Which of the following is most likely to happen when blood becomes more alkaline than normal?
(1) Oxygen will dissociate easily from oxyhemoglobin.
(2) Oxygen supply to organs will increase
(3) Respiratory rate will increase
(4) Affinity of O_2 for haemoglobin will increase

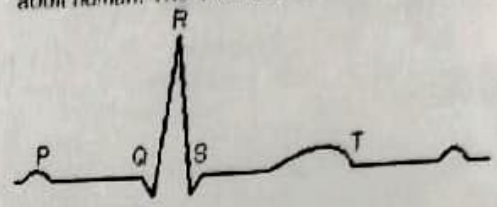
141. Plasma, an essential component of human blood constitutes around
(1) 65% of blood (2) 55% of blood
(3) 70% of blood (4) 90% of blood

142. One molecule of haemoglobin can bind maximally with _____ molecule(s) of oxygen.
Select the option which fills the blank **correctly**
(1) One (2) Two
(3) Three (4) Four

143. All of the following animals possess 3-chambered heart with two atria and a single ventricle, **except**
(1) Lizards (2) Crocodiles
(3) Snakes (4) *Chelone*

144. Assertion (A): Heart failure means the heart is not pumping blood effectively enough to meet the needs of the body
Reason (R): During heart failure, heart stops beating due to inadequate blood supply
In the light of the above statements, choose the **correct** answer from the given options
(1) (A) is false but (R) is true
(2) Both (A) and (R) are true and (R) is the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

145. The figure given below represents standard ECG of a normal adult human. The T-wave indicates



(1) Depolarisation of ventricles
(2) Excitation of atria
(3) The beginning of ventricular systole
(4) Repolarisation of ventricles

146. In humans, the major part of carbon dioxide is transported
(1) As bicarbonate
(2) As carboxyhaemoglobin
(3) As carbaminohaemoglobin
(4) In a dissolved state through blood plasma

147.

$$CO_2 + H_2O \underset{}{\overset{\text{Carbonic anhydrase}}{\rightleftharpoons}} H_2CO_3 \overset{\text{Carbonic anhydrase}}{\rightleftharpoons} HCO_3^- + H^+$$

The above reaction occurs in the backward direction at the
(1) tissue site where pCO_2 is high
(2) Alveolar site where pCO_2 is low
(3) tissue site where pO_2 is high
(4) Alveolar site where pO_2 is low

148. The approximate percentage of O_2 and CO_2 carried by plasma in dissolved state is respectively
(1) 3, 7 (2) 7, 3
(3) 3, 25 (4) 30, 70

149. The QRS complex in a standard ECG of a normal healthy individual denotes
(1) Depolarisation of the atria
(2) Depolarisation of the ventricles
(3) Repolarisation of the ventricles
(4) Repolarisation of the atria and ventricles

150. Select the **incorrect** match w.r.t. the formed elements and their percentage in a healthy individual
(1) Basophils - 2–3 per cent of total WBCs
(2) Neutrophils - 60–65 per cent of total WBCs
(3) Monocytes - 6–8 per cent of total WBCs
(4) Lymphocytes - 20–25 per cent of total WBCs

151. Exchange of gases between alveolar air and pulmonary capillaries takes place by/with
(1) Simple diffusion (2) Facilitated diffusion
(3) Active process (4) The help of proteins

152. Under normal physiological conditions the shape of oxyhemoglobin dissociation curve is
(1) Hyperbola (2) Sigmoid
(3) J-shaped (4) Ellipsoid

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153. **Assertion (A):** Blood is a connective tissue containing plasma and formed elements.
Reason (R) : It is the main circulating fluid that helps in transport of various substances.
In the light of above statements select the **correct** option.
(1) Both Assertion and Reason are true and Reason is the correct explanation of the Assertion
(2) Both Assertion and Reason are true but Reason is not the correct explanation of the Assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

154. Select the **incorrect** statement w.r.t. the nodal tissue of the heart.
(1) The entire nodal tissue, from sino-atrial node to the Purkinje fibres has the ability to generate action potentials.
(2) All parts of the nodal tissue generate the same number of action potentials per minute.
(3) Both sino-atrial and atrio-ventricular nodes are present in the right atrium.
(4) Sino-atrial node normally acts as the pacemaker.

155. Mitral valve in human heart is present between
(1) Right auricle and right ventricle
(2) Two ventricles
(3) Left auricle and left ventricle
(4) Pulmonary artery and aorta

156. Amount of O_2 delivered by 500 mL of oxygenated blood to tissues under normal physiological conditions is
(1) 100 mL (2) 500 mL
(3) 125 mL (4) 25 mL

157. Which of the following is a disorder that involves continued exposure to grinding and stone breaking industries?
(1) Asbestosis (2) SARS
(3) Flourosis (4) Common cold

158. The volume of blood pumped out by the two ventricles of human heart per minute is
(1) 5 L (2) 10 L
(3) 15 L (4) 20 L

159. Complete the analogy w.r.t. cell count present generally in per mm^3 of blood in humans.
Erythrocytes : 5 - 5.5 millions :: Thrombocytes : ____
(1) 6000 - 8000 (2) 50000 - 70000
(3) 150000 - 350000 (4) 400000 - 500000

160. All of the following together contribute almost 6-8 per cent of composition of blood plasma, **except**
(1) Globulins (2) Fibrinogens
(3) Water (4) Albumins

161. Human heart is
(1) Myogenic because it is composed of autoexcitable nodal tissues
(2) Neurogenic because ANS can regulate heart beat
(3) Myogenic because it is made of voluntary, cardiac muscles
(4) Neurogenic because it has an extensive nerve supply

162. What is the partial pressure of oxygen and carbon dioxide in the systemic artery respectively?

	Oxygen	Carbon dioxide
(1)	40 mm Hg	45 mm Hg
(2)	95 mm Hg	40 mm Hg
(3)	104 mm Hg	40 mm Hg
(4)	45 mm Hg	95 mm Hg

(1) (1) (2) (2)
(3) (3) (4) (4)

163. The gaseous product of cellular respiration is carried by haemoglobin as
(1) Carbamino-haemoglobin (2) Oxyhaemoglobin
(3) Methaemoglobin (4) Carboxyhaemoglobin

164. A special neural centre that can moderate the cardiac activity through autonomic nervous system is present in
(1) Cerebrum (2) Pons
(3) Medulla oblongata (4) Hypothalamus

165. Right shifting of oxyhemoglobin dissociation curve occurs due to
(1) Increase in pH of blood
(2) Decrease in pH of blood
(3) Decrease in partial pressure of carbon dioxide
(4) Decrease in temperature

166. The blood flows freely into the ventricles from atria during all the following events **except**
(1) Atrial systole (2) Ventricular systole
(3) Ventricular diastole (4) Joint diastole

167. Reduction in the number of which blood component can lead to clotting disorders?
(1) Lymphocytes (2) Erythrocytes
(3) Thrombocytes (4) Neutrophils

168. Match column I with column II.

	Column I		Column II
(a)	Fishes	(i)	Both pulmonary and systemic circulations are present
(b)	Amphibians	(ii)	Most of them contain 3-chambered heart
(c)	Reptiles	(iii)	Both oxygenated and deoxygenated blood mix in a single ventricle
(d)	Birds	(iv)	Heart pumps out deoxygenated blood

Choose the **correct** option.
(1) a(iv), b(iii), c(i), d(ii) (2) a(iv), b(ii), c(i), d(iii)
(3) a(iv), b(iii), c(ii), d(i) (4) a(i), b(ii), c(iii), d(iv)

169. Blood from intestine is carried to the liver via _____ before it is returned to systemic circulation. The **correct** option which fills the blank is
(1) Hepatic vein (2) Hepatic portal vein
(3) Hepatic artery (4) Inferior mesenteric artery

170. Name the chronic respiratory disorder caused mainly by cigarette smoking.
(1) Asthma (2) Respiratory acidosis
(3) Emphysema (4) Respiratory alkalosis

171. Number of double circulations completed by the human heart per minute under resting conditions is
(1) 36 (2) 72
(3) 144 (4) 18

172. Which among the following set of WBCs are agranulocytic in nature?
(1) Monocytes and Basophils
(2) Neutrophils and Monocytes
(3) Lymphocytes and Eosinophils
(4) Basophils and Eosinophils

173. Select the **odd** one w.r.t. presence of open circulatory system.
(1) Cockroach (2) Silkworm
(3) Earthworm (4) Butterfly

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174. Select the **correct** option w.r.t. first heart sound 'lub'.
(1) It is produced due to closure of atrio-ventricular valves during ventricular systole.
(2) It is produced due to closure of semilunar valves during ventricular diastole.
(3) It is produced due to closure of atrioventricular valves during ventricular diastole.
(4) It is produced due to opening of semilunar valves during atrial systole.

175. How many factors given in the box below favour the formation of oxyhaemoglobin in the alveoli?

High pO_2, Low pCO_2, High H^+ ion centration, High temperature

Choose the **correct** option.
(1) One
(2) Two
(3) Three
(4) Four

176. Read the following statements carefully and select the **incorrect** option.
(1) CO_2 combines with haemoglobin at tissue level to form carbamino-hemoglobin.
(2) CO_2 is removed from blood at the alveolar level where pO_2 is low and pCO_2 is high in lungs.
(3) Zinc is a co-factor of enzyme carbonic anhydrase.
(4) Every 100 mL of oxygenated blood delivers around 5 ml of O_2 to tissues at rest under normal condition.

177. Erythroblastosis foetalis can be avoided by
(1) Injecting Rh antigen into mother's blood immediately after delivery of the first child
(2) Administering anti-Rh-antibodies to the newborn
(3) Administering anti-Rh antibodies to mother's blood immediately after delivery of Rh positive child
(4) Injecting Rh positive blood group into mother's blood just after conception.

178. Read the given statements:
Statement A : Individuals with blood group AB are called universal recipients.
Statement B : Individuals with AB blood group harbour anti-A, anti-B antibodies in their plasma.
Choose the **correct** option.
(1) Both statements A and B are incorrect
(2) Statement B explains statement A
(3) Statement B is correct but does not explain statement A
(4) Statement A is correct

179. How many layers majorly contribute in the formation of a respiratory membrane in lungs of humans?
(1) One
(2) Three
(3) Two
(4) Six

180. Which of the following converts fibrinogens of blood into fibrins during cascade of reactions of blood clotting?
(1) Thrombin
(2) Thromboplastin
(3) Prothrombin
(4) Thromboplastin

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